

HYBRID FUSION BASED APPROACH FOR MULTIMODAL EMOTION RECOGNITION WITH INSUFFICIENT LABELED DATA

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ABSTRACT

In this paper, a deep learning based fusion approach has been proposed to classify the emotions portrayed by image and corresponding text into discrete emotion classes. The proposed method first implements intermediate fusion on image and text inputs and then applies late fusion on image, text, and intermediate fusion's output. We have also come up with a way to handle the unavailability of labeled multimodal emotional data. We have prepared a new dataset built on Balanced Twitter for Sentiment Analysis dataset (B-T4SA) dataset containing an image, text, and emotion labels, i.e., 'happy,' 'sad,' 'hate' and 'anger.' The emotion recognition accuracy of 90.20% has been achieved by the proposed method. Along with multi-class emotion recognition, we've also compared the sentiment classification results and found the proposed method to perform better than the benchmark approaches.

Index Terms— Emotion Recognition, Image and Text Emotion, Information Fusion, Intermediate and Late Fusion.

1. INTRODUCTION

With the rapid growth of multimedia data in recent years, multimodal information processing has drawn the prompt attention of the researchers [1]. Multimodal emotion processing has emerged out as a significant research trend over the last few years. Humans reflect various emotions during their communication via visual, textual, and other modalities [2]. Combining complementary information from images and texts could increase emotion recognition accuracy and help the machines become empathetic [3]. Multimodal emotion recognition has a wide range of applications such as human-computer interaction, automated identification, intelligent toys, security, service delivery, and language translation [4].

Several attempts have been made to recognize the emotional content portrayed in visual and textual modalities. In the context of text emotion recognition, X. Glorot et al. [5] attempted large-scale text sentiment analysis using deep neural networks along with domain adaptation techniques. Image Emotion Recognition is also an equally active research area.

In this regard, T. Rao et al. [6] proposed a new deep network to prepare hierarchical representations for emotion recognition in visual domain. Multimodal emotion recognition has drawn the attention of researchers in the last few years. For instance, P. Fortin et al. [7] implemented a multitask architecture based emotion recognition approach to perform predictions with one or two missing modalities by using a classifier for each combination of image, text, and tags. In another work, Nan Xu et al. [8] modeled the interplay of visual and textual content for sentiment recognition using co-memory based network.

Unavailability of sufficient labeled multimodal data annotated with various emotion classes is also a significant challenge faced by multimodal emotion recognition. Most of the available datasets and literature focused on classifying emotions portrayed by multimodal data into positive, negative, and neutral classes. For example, D. Onita et al. [9] proposed an approach for mapping images to text for sentiment classification. In another work, S. Poria et al. [10] presented a novel method to classify emotions using multiple kernel learning classifier that used the features from image and text domains. N. Xu et al. [11] proposed a multiview framework for sentiment analysis consisting of a deep semantic network for sentiment classification based on the object, scene, and text. Multi-class emotion recognition has not been fully explored for multimodal emotion data. It inspired us to compile a multimodal dataset labeled with discrete emotion classes, i.e., 'happy,' 'sad,' 'hate,' and 'anger,' and work towards multi-class emotion recognition.

The proposed system encompasses intermediate and late fusion phases. Intermediate fusion is first implemented on image and text input, and then late fusion is applied to IER module's output, TER module's output, and intermediate fusion's output. We have implemented separate image emotion recognition (IER) and text emotion recognition (TER) models on the image and text samples of the Balanced Twitter for Sentiment Analysis (B-T4SA) dataset [12]. The samples with the same predicted emotion label for both text and image were used to build the multimodal emotion dataset. The proposed method has achieved an emotion recognition accuracy of 90.20%. The code and dataset access is available at github.com/MIntelligence-Group/Image-Text_EmoRec.

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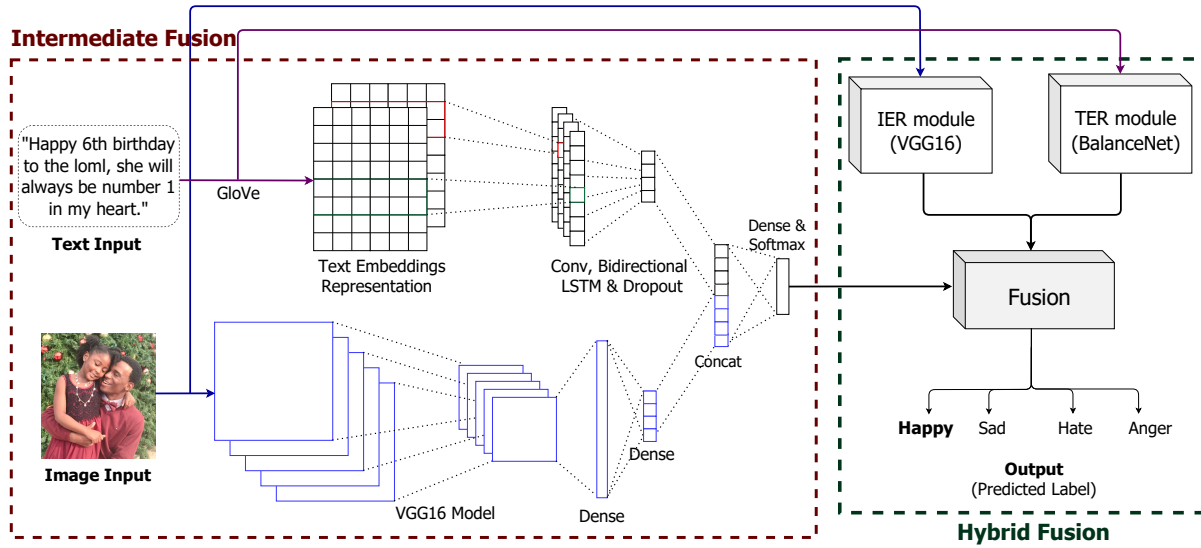


Fig. 1: Schematic architecture of proposed method. Here, IER: Image Emotion Recognition, TER: Text Emotion Recognition.

The contributions of the paper are as follows.

- A novel deep learning based fusion approach has been proposed for multimodal emotion recognition. The proposed method classifies a data-sample containing an image and corresponding text into discrete emotion classes.
- We have also come up with a way to handle labeled multimodal emotional data's unavailability. We have prepared a new dataset containing an image, corresponding text, and emotion labels ('happy,' 'sad,' 'hate,' and 'anger').
- Ablation study and extensive experiments have been carried out. The proposed approach has demonstrated better performance than the state-of-the-art methods for sentiment classification as well.

The rest of the paper is organized as follows. Section 2 describes the proposed method. Experiments and results are discussed in Section 3. Section 4 concludes the paper and highlights future research directions.

2. MULTIMODAL EMOTION RECOGNITION

This section discusses the proposed approach for multimodal emotion recognition from images and corresponding text. Its architecture is described in Fig. 1, and it is selected as per the ablation study discussed in Section 3.1.3.

2.1. Dataset Preparation

Most of the datasets available for image and text contain sentiment labels, i.e., positive, negative, and neutral [13, 14, 15]. We implemented separate image emotion recognition (IER) and text emotion recognition (TER) models on the image and text samples of the B-T4SA dataset. The data-samples

for which the same emotion class label was predicted by both TER and IER models were included to prepare the multimodal dataset for multi-class emotion recognition. The dataset thus prepared is described in Fig. 2.

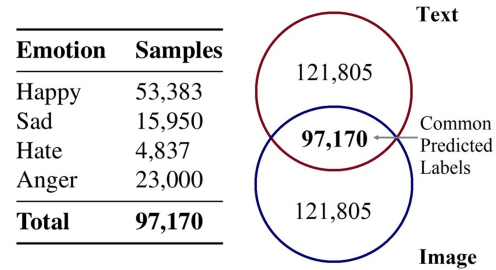


Fig. 2: Dataset summary. Left: number of examples for each emotion class. Right: number of examples in each modality.

The detailed architectures of IER and TER modules have been discussed in Section 2.2. The IER module was trained on Flickr Image Dataset [16]. The image data-samples labeled as 'excitement' were re-labeled as 'happy.' The samples labeled 'disgust' were re-labeled as 'hate.' The TER module was trained on Twitter Text Dataset [17]. Text pre-processing was carried by making all letters lowercase, removing punctuation and stop-words from it. Lemmatization and correction of letter repetitions were also performed. To balance the Twitter dataset, we added 3000 and 1000 data-instances of class 'anger' and 'hate,' respectively. The samples labeled as 'worry' were merged into the samples labeled 'sad.' While the samples labeled as 'enthusiasm,' 'surprise,' 'love,' 'fun,' 'happiness,' and 'relief' were re-labeled as 'happy.' The merging of emotion labels has been done based on Plutchik's wheel of emotions [18].

2.2. Intermediate Fusion Phase

Intermediate fusion combines information from text and image modalities before attempting to classify the emotion class. The IER module used a pre-trained VGG16 model [19]. We added two dense layers of dimensions 2000 and 500, respectively, followed by a flatten layer. Image encoding vector I_e was obtained as the output. The TER module extended BalanceNet [20] for multi-class emotion recognition instead of sentiment classification. The pre-trained BalanceNet takes GloVe representations [21] of the text as input, and it has two channels. These channels had a combination of convolution layer and a dropout layer with a dropout value of 0.3, along with a bidirectional LSTM layer. We added the flatten layers to both the channels and obtained the text encoding vector T_e . Further, I_e and T_e were concatenated as per Eq. 1. The output of the concatenation layer was then fed to a dense layer of dimension 1000 and then to a dense layer of size 4 with softmax as the classifier.

$$y_i = \text{softmax}(\text{concat}(I_e, T_e)^T W + b) \quad (1)$$

$\left\{ \begin{array}{l} \text{where :} \\ W : \text{Weight matrix and } b : \text{Bias term} \end{array} \right.$

2.3. Late Fusion Phase

Late fusion classifies the emotion for each modality before attempting to fuse the results. We first found the predicted probabilities for four emotion classes, i.e., ‘happy,’ ‘sad,’ ‘hate’ and ‘anger’ for the following three modalities - image only, text only, and intermediate fusion of image and text. Then the outputs of the IER model, TER model, and intermediate fusion were fed as input to the late fusion phase. Further, the weight coefficients for these modalities were computed using a grid search, and the weighted average was performed to find the final predicted label as per the following steps.

i) Initialize the possible choices for the coefficient weights as $\text{weight}_{ch} = [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]$.

ii) Compute the coefficient weights as per Algorithm 1. The experimentally observed coefficient weights for TER, IER and intermediate fusion modules are 0.4737, 0.2632 and 0.2632.

Algorithm 1: Grid search to find coefficient weights

Result: W_f
Define $W_i = \text{random}(\text{weight}_{ch})$ where $i \in \{1, 2, 3\}$;
Define $W_f = \text{final coefficient weights}$;
Define $\text{score} = 0, \text{score}_{best} = 0$;
Define $\text{acc_score}() = \text{function to compute score}$;
Define $p_{ij} = \text{predicted probability by } i^{\text{th}} \text{ modality}$;
 $\text{score} = \text{acc_score}(\text{argmax}(\sum_{i=1}^3 (p_{ij} * W_i)))$;
if $\text{score} > \text{score}_{best}$ **then**
 | $\text{score} \leftarrow \text{score}_{best}$;
end

iii) For each emotion class, perform the weighted average of the predicted probabilities of the emotion labels and coefficient weights obtained in the previous step.

$$P_j = \sum_{i=1}^3 (p_{ij} * W_f) \quad (2)$$

$\left\{ \begin{array}{l} \text{where :} \\ i \text{ corresponds to modalities (image, text, fusion)} \\ j \text{ corresponds to labels (happy, sad, hate, anger)} \\ W_f = \text{Final coefficient weights} \\ p_{ij} = \text{Predicted probability by } i^{\text{th}} \text{ modality} \\ P_j = \text{Final probability for } j^{\text{th}} \text{ class} \end{array} \right.$

iv) Compute the final predicted label, $\hat{y}$ using Eq. 3 by finding the class corresponding to the maximum probability value.

$$\hat{y} = \text{argmax}(P_j) \quad (3)$$

3. EXPERIMENTS AND RESULTS

3.1. Implementation

3.1.1. Experimental Setup

The network proposed in this paper has been trained on Nvidia Quadro P5000 GPU with 2560 CUDA cores and 16 GB Virtual RAM. The testing of the trained network has been carried out on Intel(R) Core(TM) i7-7700, 3.70 GHz, 16GB RAM CPU system with 64-bit Ubuntu Linux OS machine.

3.1.2. Dataset and Training Strategy

The dataset preparation and training strategy of the IER and TER models has been discussed in Section 2.1. The proposed network has been trained using a validation split of 0.3. Sparse Categorical Crossentropy loss function and Adam optimizer are used. Model testing is carried out thrice on around 7,000 unseen samples, and average results are reported.

3.1.3. Ablation Study

The ablation study on the B-T4SA dataset is summarized in Table 1. Intermediate fusion fuses the information from Image & Text modalities. Late fusion-1 performs a simple fusion of Image modality and Text modality’s outputs. Late fusion-2, which is named the ‘Hybrid Fusion,’ fuses the information from Image modality, Text modality, and Intermediate fusion’s output in a hybrid manner.

Table 1: Summary of the ablation study

Model	Accuracy
IER Only	76.42%
TER Only	82.92%
Intermediate Fusion-1	27.94%
Intermediate Fusion-2	75.94%
Late Fusion-1	89.80%
Hybrid/Late Fusion-2	90.20%

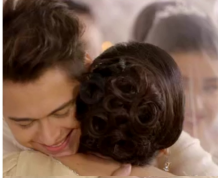
			
"Apart from SerTen, these mother-son moments really captivated my heart."	"Boys and young men can be victims of Child Sexual Exploitation"	"Don't you look at me in the mornings, like not even accidentally. I will cut you."	"This is gaza- not Hiroshima. Gaza is being further bombed by Israeli bombs today."
GT: Happy P: Happy	GT: Sad P: Sad	GT: Hate P: Hate	GT: Anger P: Anger
			
"If you're having a bad day here's a bird riding a dog like a horse."	"LOST KONIKoni was lost around 8/20 and found 8/21 but escaped from who she was caught by on 8/22"	"Anyone else thinking that this whole election is just a horribly acted social experiment?"	"I'm going to be very upset if Stranger Things season 2 is not an extended monologue from Barb"
GT: Happy P: Happy	GT: Sad P: Sad	GT: Hate P: Hate	GT: Anger P: Anger

Fig. 3: Results for sample data containing image and corresponding text. Here, ‘GT’: ground-truth, ‘P’: predicted output label.

3.2. Results & Evaluation

The proposed approach has been evaluated as follows.

3.2.1. Accuracy & Confusion Matrix

The proposed approach has resulted in an emotion recognition accuracy of **90.2%**. Fig. 4 shows the confusion matrix.

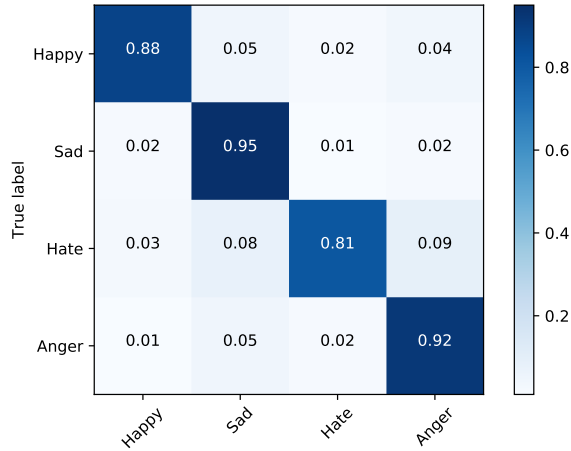


Fig. 4: Confusion matrix.

3.2.2. Comparison with State-of-the-art Methods

Table 2 compares the performance of the proposed approach with the state-of-the-art (SOTA) methods for sentiment classification on the B-T4SA dataset. The accuracy discussed here is for sentiment classification as SOTA results are available for the same. Whereas, the final accuracy discussed in section 3.2.1 is for multi-class emotion recognition.

Table 2: Result comparison with state-of-the-art methods

Model	Author	Accuracy
Cross-Media Learning	L. Vadicamo [12]	51.30%
Image-Text Sentiment Fusion	A. Gaspar [22]	60.42%
Proposed Method		86.70%

3.2.3. Sample Predictions

The above sections have shown the quantitative results while the qualitative results have been presented in Fig. 3 by showing some of the data samples and their predicted output.

4. CONCLUSION AND FUTURE WORK

The multimodal approach proposed in this paper has obtained better accuracy than IER and TER alone for multi-class emotion recognition. The proposed approach has outperformed the state-of-the-art benchmark approaches for sentiment classification. It proves the applicability of using the information from multiple modalities to improve emotion recognition. In the future, we aim to extend the emotion recognition for more emotion classes and incorporate more modalities such as speech and videos.

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